

Quindaro Ruins Land Management Plan

By

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Abstract

This Land Management Plan has been created as a capstone project for the University of Kansas Environmental Studies Program. It was developed concurrently with a companion Cultural Landscape Assessment (Brown, 2023), which should be consulted alongside this document for a complete picture of the site's cultural and historical significance. This plan is designed for the Quindaro Ruins Project Foundation and its partners, including site planners, land managers, and preservation stakeholders.

The objectives of this land management plan are to assess the Quindaro Ruins site from an ecological and archaeological standpoint. Specifically, this plan: (1) provides a brief overview of the ecology and history of the Quindaro Township; (2) identifies goals for land development; (3) assesses the site's environmental and ecological values; (4) identifies environmental threats; (5) examines the condition of ruins and structures; (6) presents options for invasive honeysuckle and vegetation removal; (7) outlines trail implementation; and (8) discusses potential for recreational park development.

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1. Overview of Project and Site

Landowner: The African Methodist Episcopal Church currently owns the majority of the lands that comprise the historic Quindaro townsite.

Location: The Quindaro townsite is located in Wyandotte county, south of the Missouri river in sections 29 and 30 in Kansas City Kansas. See Figure 1, below.

Size: The Quindaro Townsite is 70.5 acres.

Description: Founded in 1856 on Wyandot ancestral land, The Quindaro townsite is a complex, urban archaeological site with a rich history and significant ties to the Underground Railroad. It was also an example of a 19th-century African American settlement and was home to Western University, widely recognized as the first school for African Americans established west of the Mississippi River, which offered educational opportunities to formerly enslaved people and their families.

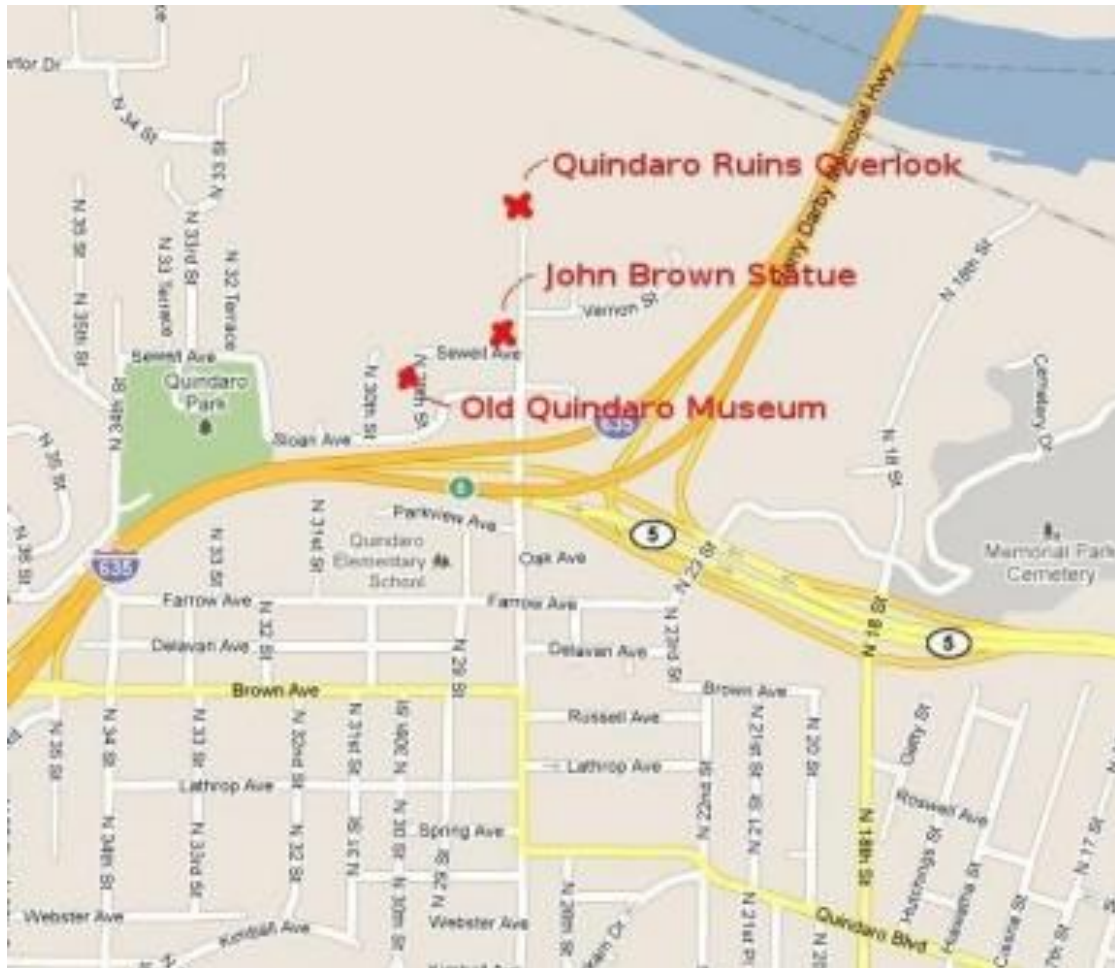


Figure 1. Quindaro & Western University Ruins (Source: kansastravel.org/kansascitykansas/quindaro.htm)

2. Conservation Significance

Environmental:

The Quindaro Ruins are located in the Kansas floodplain and on bluffs overlooking the Missouri River. Floodplains provide essential ecosystem services, as they filter and store water, help with flood control, and are integral to healthy functioning river systems. Many species of animals and plants live in floodplain areas and adjacent riparian corridors and bluffs. The soil of floodplains is notably fertile, supporting agriculture and diverse plant communities.

Human development and industrialization on floodplains and within river corridors contribute to increased risks with flooding, erosion, loss of biodiversity, and other environmental degradations. Quindaro's floodplain setting offers significant potential for low-impact recreation, including hiking, fishing, and biking, while providing opportunities to connect visitors with the natural landscape. Any development should minimize environmental disturbance and, where possible, restore floodplain and riparian corridor vegetation.

Figure 2 illustrates the ecological value at the Quindaro Ruins site, using raster data set from the "2018 Green Infrastructure analysis for Kansas City region" created by the Mid America Regional Council. The map demonstrates a positive relationship between proximity to the Missouri River and ecological value. Higher ecological value is also associated with greater tree canopy density. A small stream and creek on-site also exhibit elevated ecological value in their immediate surroundings. Notably, areas cleared of vegetation show a direct correspondence with lower ecological value — an issue discussed further in relation to trail installation.

Ecological value at Quindaro Ruins

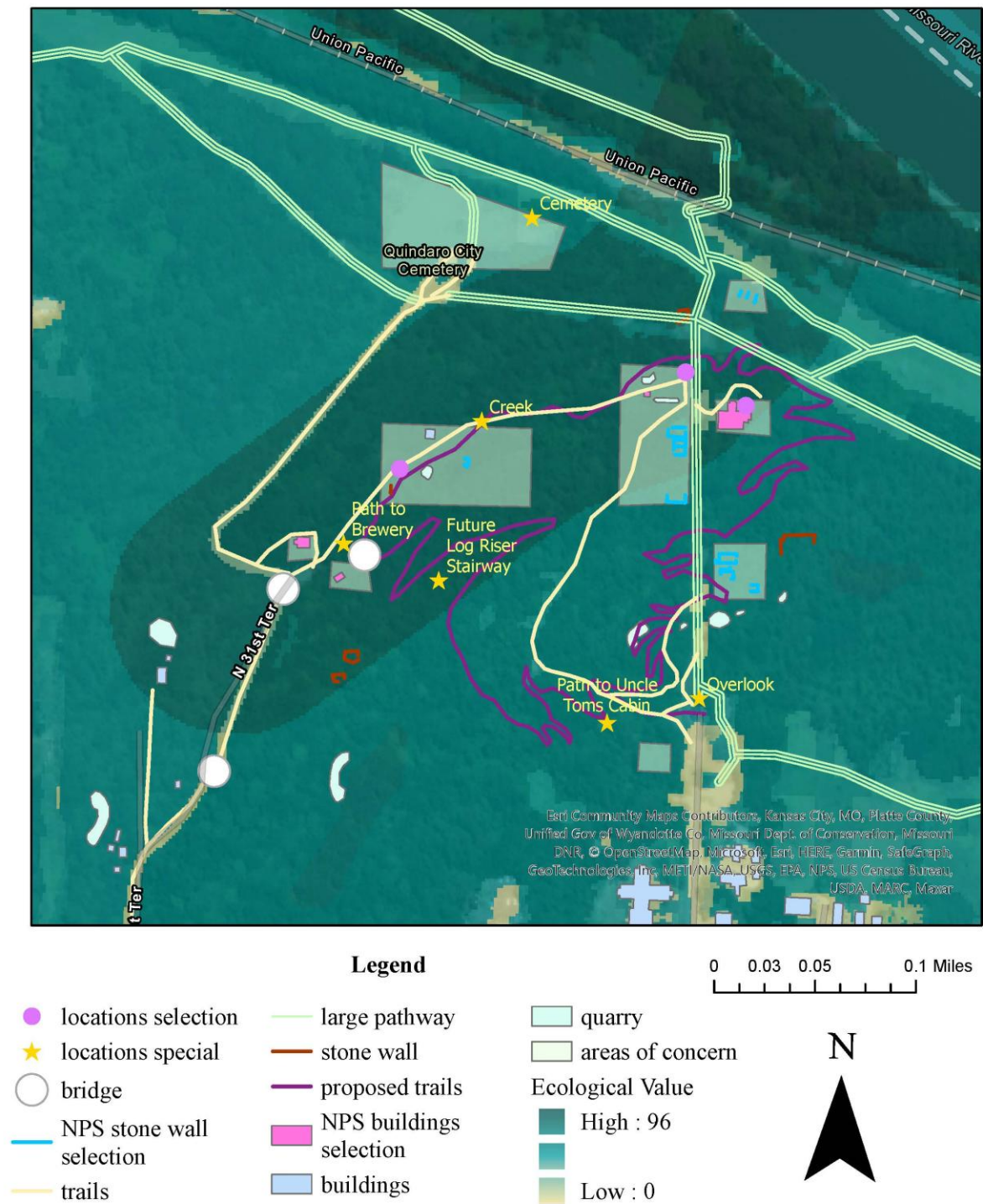


Figure 2. Ecological Value at Quindaro Ruins (created by Kaitlyn Rawlings with data from MARC (2023))

Cultural:

The Quindaro Townsite holds significant importance in Black American history. It served as a prominent station on the Underground Railroad network and is an important site for U.S. Civil War history. Enslaved people would escape from Platte County, Missouri to Quindaro, where they would shelter with local farmers before continuing toward Nebraska and freedom. As many inhabitants were recruited to join the Union Army during the Civil War, the lower half of the township was largely abandoned. The upper half remained predominantly inhabited by African Americans and formerly enslaved people. See Figure 3 below for a historic plat of the original town.

Quindaro was also home to Western University. Western University was founded in the 1860s by Eben Blachly. The school educated the children of escaped slaves and other black families in the area, making it the first school for African Americans west of the Mississippi River. Western University served this community for decades until it was heavily impacted by the Great Depression, resulting in its closure in 1940. Today, the remaining physical traces of the school include the John Brown statue and several cornerstones of the university building in Quindaro, Kansas City. For a more thorough treatment of the site's cultural history, refer to Tamar Brown's Cultural Landscape Assessment (Brown, 2023).

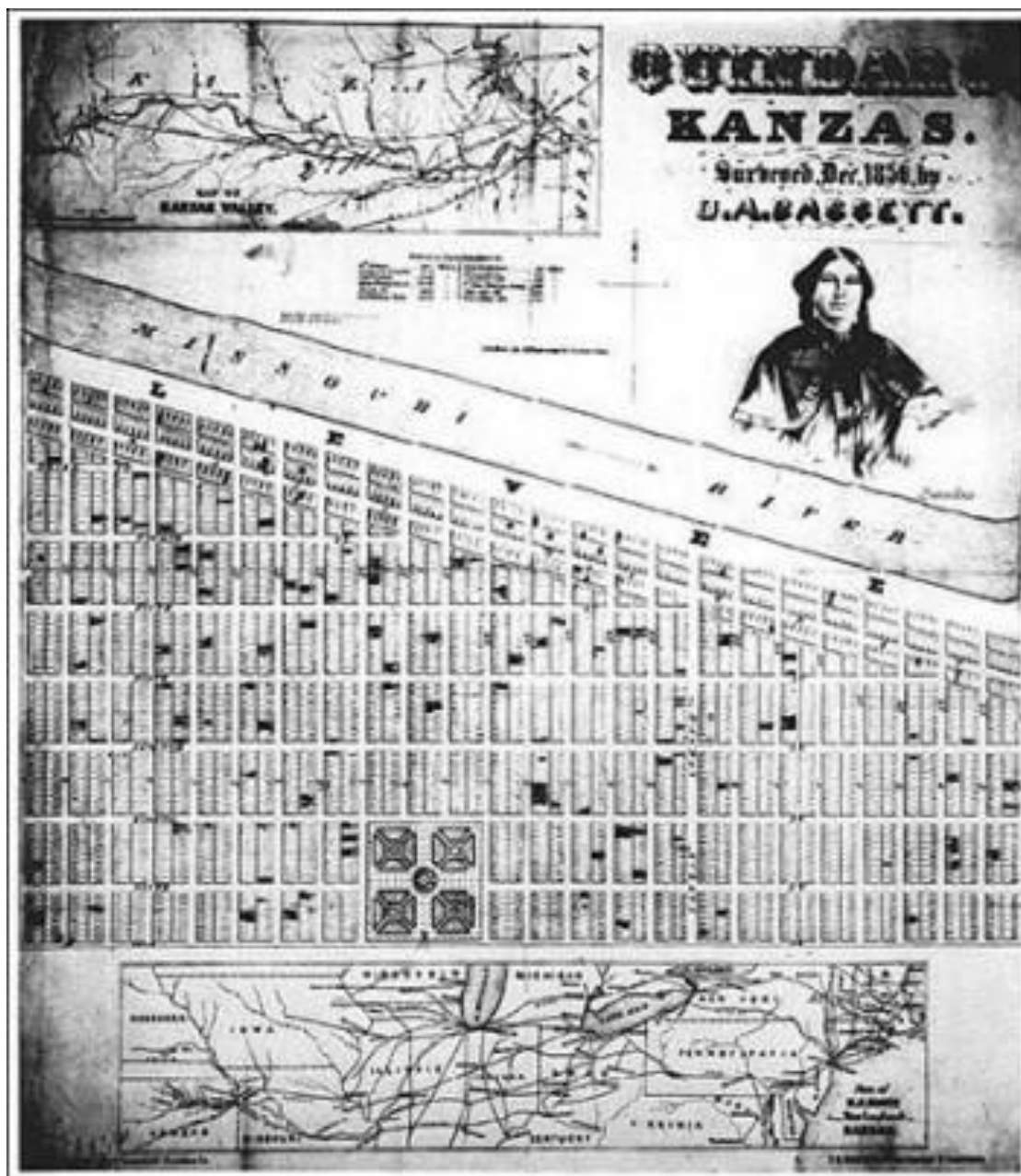


Figure 3. Historical Map of Quindaro Township (Kansas City Kansas Public Library, kckpl.org/kansas/quindaro.html)

3. Vegetation

Friends of Kaw Point Park has documented many native plant species present on its grounds. Kaw Point Park is 4.6 miles away from the Quindaro Ruins. Given it's proximity much of these species are likely to be native to the Quindaro site as well. Additionally, Tamar Brown's companion Cultural Landscape Assessment (Brown, 2023) identifies flora documented in Wyandotte County during the period when Quindaro was an active settlement. Both data sources are presented in Figure 4 below.

Current Native Vegetation at Kaw Point Park (Kaw Point Park, 2023)	Historic Vegetation in Wyandotte County (Brown, 2023)
- Columbine - Aromatic aster - Black walnut - Big bluestem - Cottonwood - Elderberry - Fragrant sumac - Illinois bundleflower - Pawpaw - Peach-leaved willow - Plains prickly pear - Prairie blazingstar - Plains coreopsis - Purple coneflower - Rough blazingstar - Serviceberry - Showy partridge pea - White sage - Wild ginger - Wild plum	Major Tree Species: - Bitternut hickory - Shagbark hickory - White oak - Red oak - Black oak - Burr oak - Sycamore - Mulberry - Cottonwood - Bass wood - Walnut - White elm - Red elm - Hackberry - Kentucky coffeetree - Ironwood - Ash Vines: - Morning glory - Silk vine - Wild cucumber - Pea vine - Hop - Bittersweet - Moonflower Shrubs & Bushes:

	Wild rose Red bud Dogwood Wild currant Wahoo Buckeye Prickly ash Sumac Indian cherry Hop tree Herbaceous Species: Columbine Purple coneflower Wild ginger Fruits & Nuts: Walnuts Hickory nuts Hazelnuts Persimmon Pawpaw Wild grapes Blackberries Strawberries Raspberries Gooseberries Plums Crabapple Red/Black haws Dewberries Wild cherries Elderberries Service berries
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Figure 4. Table of Current and Historic Vegetation in Wyandotte Area (Sources: Kaw Point Park (2023); Brown

4. Project Goals

Overall Management Goals: The overall management goals for the Quindaro Ruins site are:

- Stabilize the ruins and protect their archaeological integrity during and after any development activities.
- Minimize environmental disturbance during site development, and actively eliminate ecological threats — particularly invasive Asian bush honeysuckles (*Lonicera maackii*, *L. tatarica*, *L. morrowii*).
- Restore the landscape to a healthier ecological condition, which also helps suppress invasive species over the long term.

These three goals are interdependent. To protect archaeological integrity requires managing vegetation, and ecological restoration adds to the long-term viability of the ruins and future infrastructure.

5. Ecological Concerns

Ecological Threats:

The primary ecological threats to the Quindaro Ruins site are vegetation overgrowth and invasive Asian bush honeysuckles (*Lonicera maackii*, *L. tatarica*, *L. morrowii*). These species outcompete native plants for resources and accelerate the deterioration of archaeological structures.

Structural Concerns:

Two maps (Figures 5 and 6) display the Depositional Integrity and Quality of Conditions of the 23 NPS assessment features at Quindaro. ‘Depositional Integrity’ refers to a feature’s degree of archaeological preservation, while ‘Conditions’ refers to its current physical state. Features labeled as *Poor* should be prioritized first from a stabilization perspective.

Information below is summarized from Jacob Moody’s *Condition Assessment of Archeological Features at Historic Quindaro, Kansas City, KS* (National Park Service, U.S. Department of the Interior, 2022).

Depositional Integrity Rankings:

Good: shows no evidence of noticeable deterioration by natural forces and/or human activities. The feature is considered currently stable, and its present archeological values are not threatened. No adjustments to the currently prescribed feature treatments are required, in the near future, to maintain the feature's present condition.

Fair: shows evidence of deterioration by natural forces and/or human activities. If the identified impacts continue without the appropriate corrective treatment, the feature will degrade to a poor condition and the feature's data potential for historical or scientific research will be lowered.

Poor: shows evidence of severe deterioration by natural forces and/or human activities. If the identified impacts continue without the appropriate corrective treatment, the feature is likely to undergo further degradation and the feature's data potential for historical or scientific research will be lost.

Inundated/Uncertain: The deposits and condition of an inundated feature, formerly in a terrestrial setting, are obscured and cannot be accurately assessed due to factors such as water turbidity or natural lack of clarity, wave action, growth of aquatic vegetation, and other conditions. Application of standard methods to assess the condition of an inundated feature is not possible in these circumstances.

Not Relocated/Unknown: The location where the feature was last documented was visited, but the feature could not be relocated. Based on best professional judgment that considers standard feature types in the site, the area is deemed to most likely be the location of the feature. Further testing may be required to determine the feature location.

Destroyed: The feature's formal condition assessment resulted in a professional determination that the feature was destroyed or so severely damaged that the data potential/scientific research value was deemed insufficient to warrant further archeological monitoring or investigation.

Conditions Rankings:

The Conditions ranking system uses identical categories and definitions to those described above for Depositional Integrity. See Section 5b for full definitions of Good, Fair, Poor, Inundated/Uncertain, Not Relocated/Unknown, and Destroyed.

Depositional Integrity on NPS Assessment Features

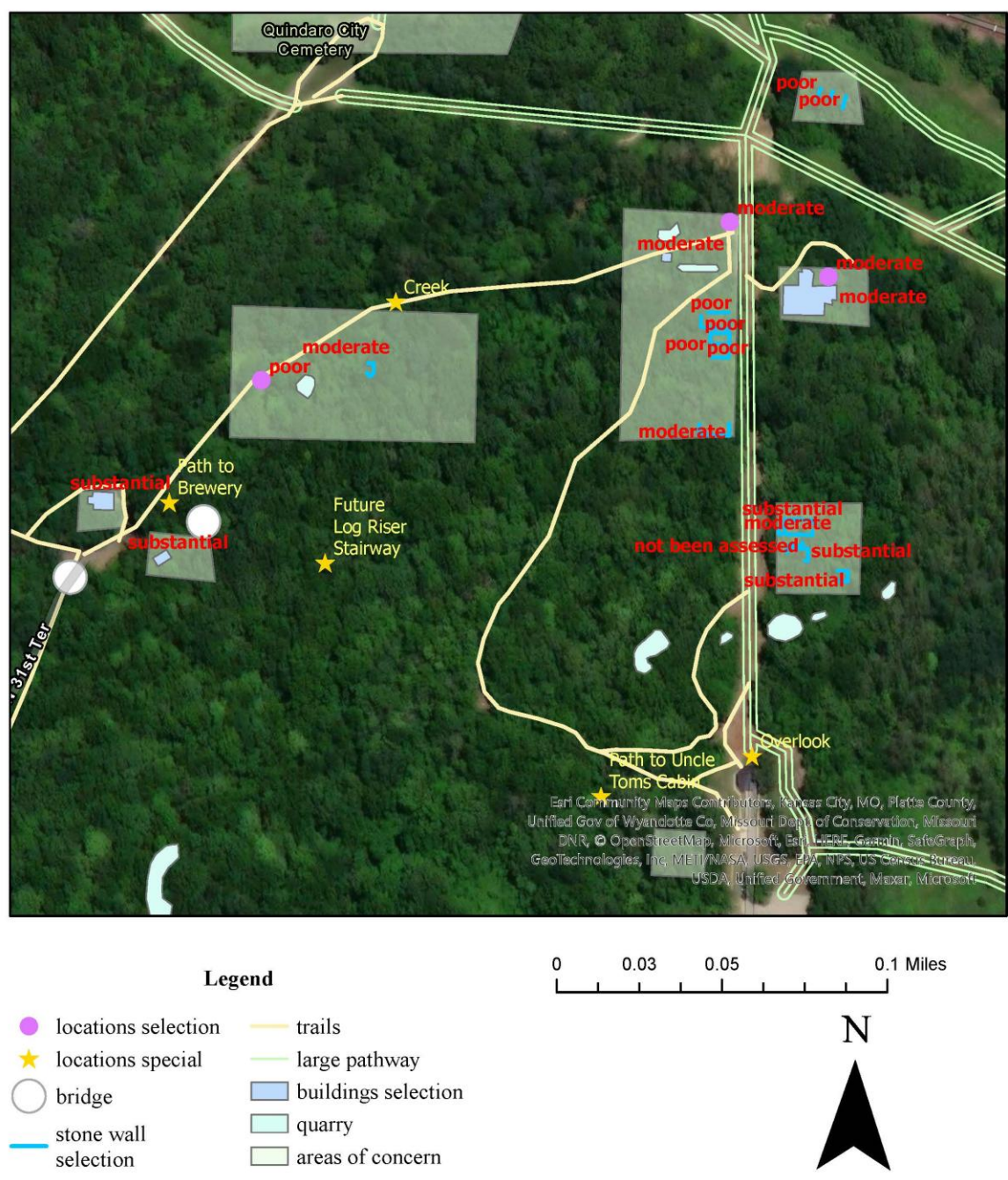


Figure 5. Depositional Integrity of NPS Assessment Features (created by Kaitlyn Rawlings with data from National Park Service (2022))

Quality of Conditions on NPS Assessment Features

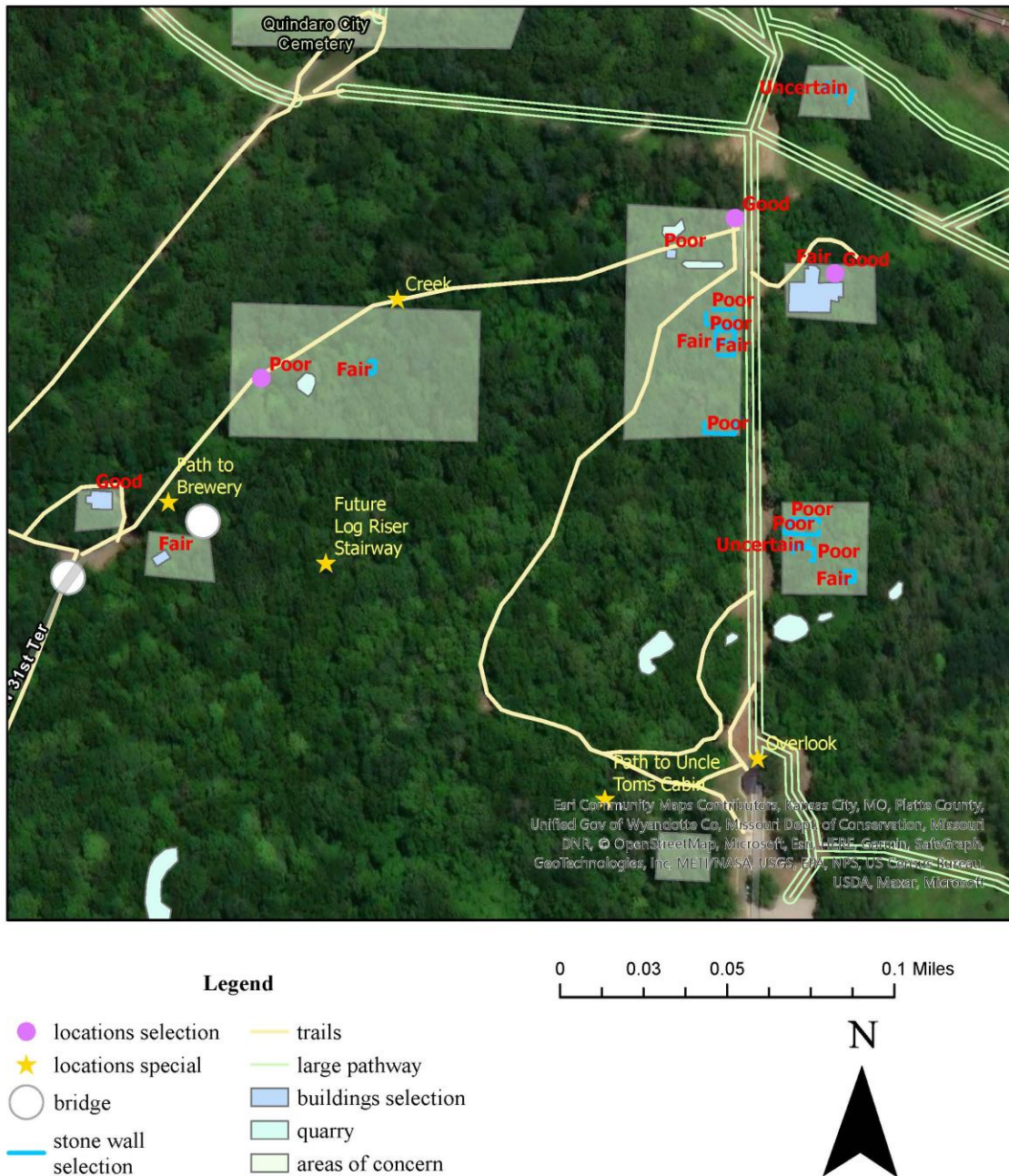


Figure 6. Quality of Conditions on NPS Assessment Features (created by Kaitlyn Rawlings with data from National Park Service (2022))

6. Management Plan

Vegetation Removal around Structures:

Vegetation, if left unmonitored, increases the degradation of manmade and archaeological structures. Many of the archaeological structures in Quindaro are covered in an overgrowth of plants, resulting in their further loss of integrity and conditions. For this reason, it is important to remove the vegetation surrounding these sites. Where tree root systems may be actively supporting the structures, complete removal should be avoided. Instead, felling (cutting the trunk while leaving roots intact) may need to be implemented, to minimize below ground disturbance. Plants with shallow roots not supporting the structure can be removed completely.

Most of the archeological structures in Quindaro need some sort of vegetation removal, and several are severely overgrown by vegetation, particularly shrub honeysuckle. Below in Figure 7 is a map of areas in Quindaro ruins that should be prioritized in vegetation removal. NPS did an assessment on 23 features in Quindaro and the features highlighted in orange are the ones suffering the most from overgrowth of vegetation.

Prioritized removal of overgrowth on NPS Assessment Features

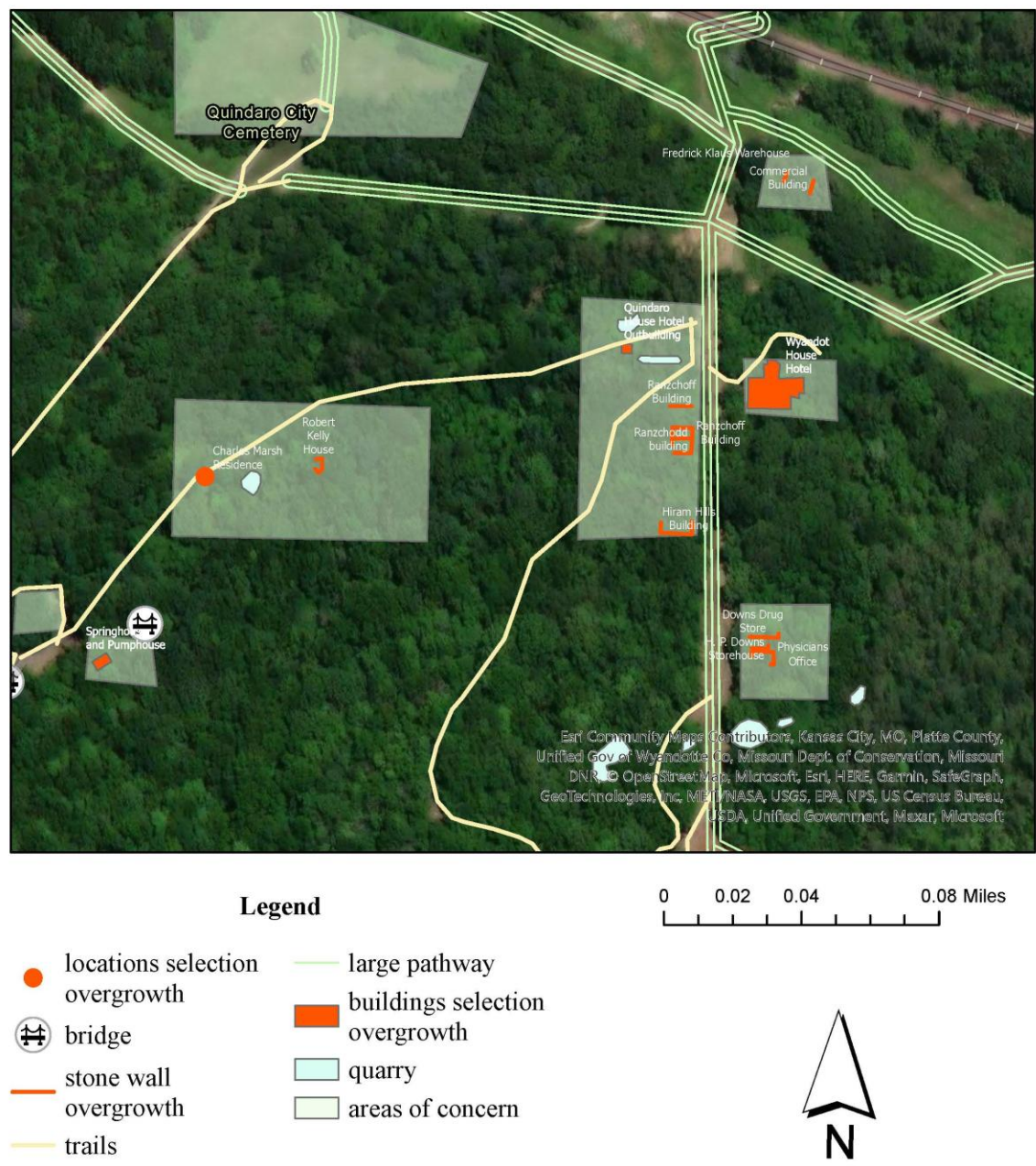


Figure 7. Prioritized Removal of Overgrowth on NPS Assessment Features (created by Kaitlyn Rawlings with data from National Park Service (2022))

Invasive Honeysuckle Management:

The Quindaro Ruins land contains large expanses of Asian bush honeysuckles (*Lonicera maackii*, *L. tatarica*, *L. morrowii*). These are an invasive species native to Asia that are commonly found on floodplains and woodland edges in the eastern third of Kansas. These plants outcompete native vegetation for resources and reduce the structural integrity of the ruins, as well as accelerate their deterioration.

There are several options for the removal of honeysuckle. These options fall into two categories, mechanical and chemical. There has previously been mechanical removal of honeysuckle in the Quindaro Ruins by volunteers organized by Johnson County Community College (removing the honeysuckle by hand); however, there are still large expanses of honeysuckle throughout the park. A combined approach of chemical and manual removal is recommended for this site.

The honeysuckle removal plan is organized into phases with Phase 1 focusing on prioritized removal of overgrowth on NPS features along with a 30-foot buffer zone around each prioritized feature to prevent regrowth. The buffer zone will expand by an additional 30 feet until the removal areas merge. This plan is depicted below in Figure 8.

Quindaro Ruins Honey Suckle Removal Phases

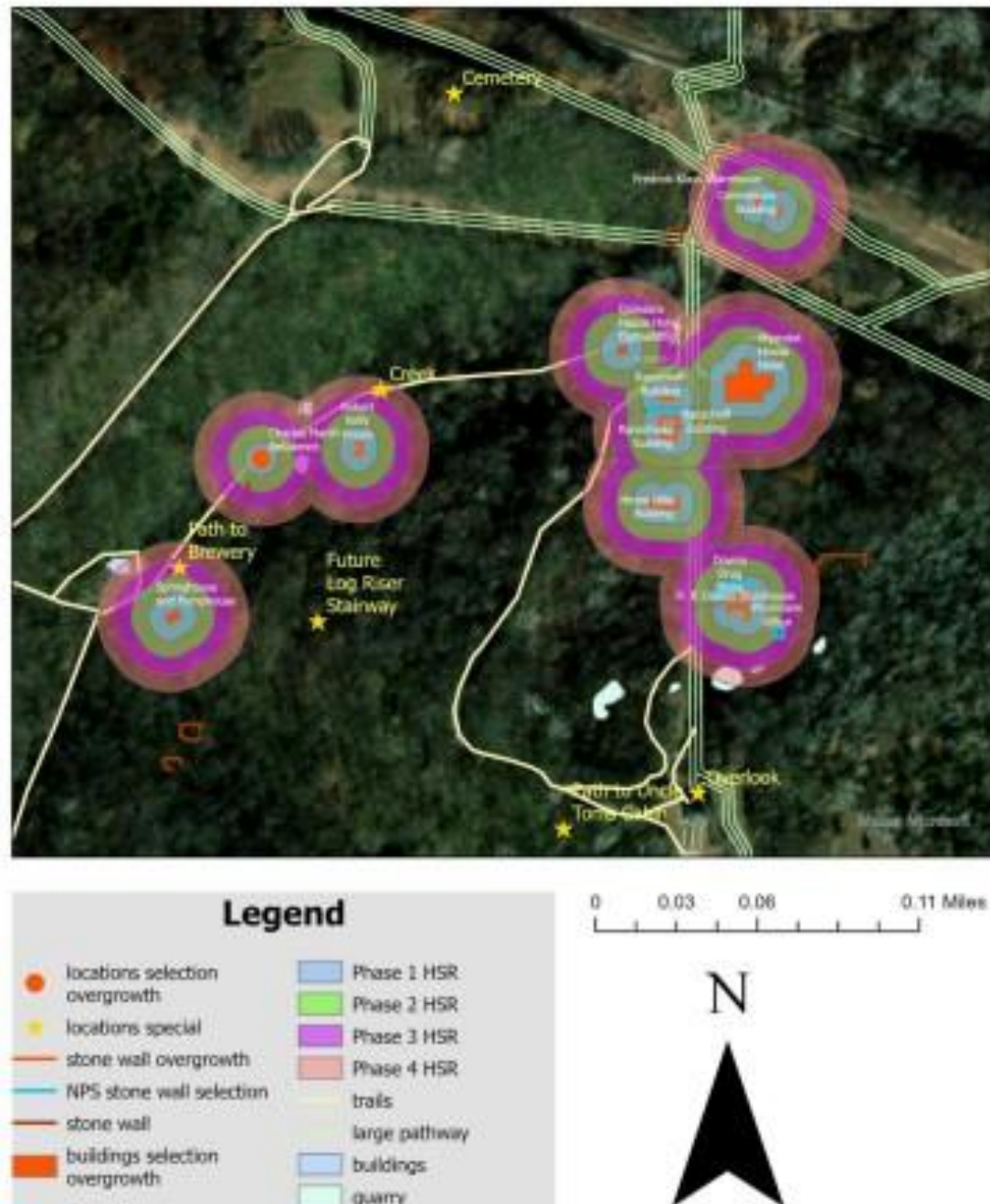


Figure 8. Quindaro Ruins Honeysuckle Removal Phases (created by Kaitlyn Rawlings)

i. Manual Removal

For manual removal, shrubs are cut off at the base using loppers or hand saws. The cut stump should be immediately sprayed with a low-pressure sprayer or dabbed with Tordon RTU (Picloram, 2,4-dichlorophenoxyacetic acid) or a 20% solution of glyphosate (Roundup or Rodeo) to prevent regrowth. Treatment should be targeted to the stump only to protect nearby native vegetation or conducted in fall after other species have gone dormant.

Follow-up spot-spraying of regrowth with an 8–10% glyphosate solution is recommended in spring before other species emerge, or in fall after dormancy. Manual removal is best suited to smaller, discrete patches of honeysuckle.

ii. Chemical Removal

A mist-sprayer technique can treat large expanses of honeysuckle efficiently, at low cost per acre, covering 1–2 acres per hour. Misting must be conducted in late fall when native plants are dormant; this window in Kansas is late October to mid-November. The mist can reach approximately 15 feet. Given the extent of honeysuckle at Quindaro, a mist application is well suited for this site. Because of Quindaro's high ecological value, this plan recommends using a backpack mist blower to implement misting, as opposed to making alleys every 15 feet. The Kansas Forest Service has backpack mist blowers available for loan — contact Ryan Armbrust at rarmbrust@ksu.edu to request equipment.

iii. Goats

Goat grazing has been employed successfully by the City of Lawrence and other Kansas municipalities to control honeysuckle. Goats naturally prefer sweeter-tasting foliage and tend to avoid bitter native plants, selectively targeting honeysuckle. Their droppings also provide natural fertilization beneficial to native species. Several Kansas cities have contracted *Goats On The Go* for this purpose. This option may be particularly suitable for areas where chemical use is undesirable or restricted.

Revegetation:

Following honeysuckle removal, revegetation is essential to reestablish ground cover and reduce the risk of reinvasion. The historic plant communities of the area, documented in Figure 4, should guide species selection for revegetation, with priority given to natives appropriate to the site's floodplain and upland bluff conditions. Appropriate groundcover such as native cool season grasses and understory forbs and wildflowers should be seeded. Native understory shrubs and trees that will outcompete the honeysuckle should be planted from seedlings.

The following partners should be consulted:

- Friends of Kaw Point Park — revegetation experience and species recommendations
- K-State Extension
- Deep Roots KC — native species identification, resources, and restoration protocols

A native species showcase area should be established near the overlook along the trail (see Figure 9), with interpretive signage about the historic landscape, invasive species management, and restoration ecology. A native species arboretum along the trails could highlight existing high-quality native trees and vegetation groupings as described in Brown (2023).

A formal, phased revegetation plan should be developed following completion of a site vegetation survey (see Section 8).

7. Trails Implementation Plan

Trail development at Quindaro presents an opportunity to increase public access to a historically and ecologically significant site but requires careful siting to minimize environmental impact. Any trail installation inherently fragments the landscape, and these edge effects — reduced population density, lower biodiversity, and disrupted community structure — must be weighed seriously when planning routes. Two case studies examining trail-related biodiversity impacts are linked in the Useful Links section.

Because Quindaro is a site of high ecological value, a revised trail alignment is proposed (shown in blue on Figure 9). The revised route builds on the previously proposed path while incorporating several improvements: where possible, new segments are aligned with already-existing trails at the site, reducing the amount of new ground that must be disturbed. In areas of particularly high ecological value, the trail was redrawn to avoid creating unnecessary edges in the landscape. This approach also makes practical economic sense — the more the plan leverages existing infrastructure, the less new trail will need to be installed.

Trail design should follow low-impact best practices, and any new trail segments should be evaluated for soil stability, drainage, and proximity to sensitive archaeological and ecological features before construction begins.

Proposed Bike Trails at Quinaro Ruins

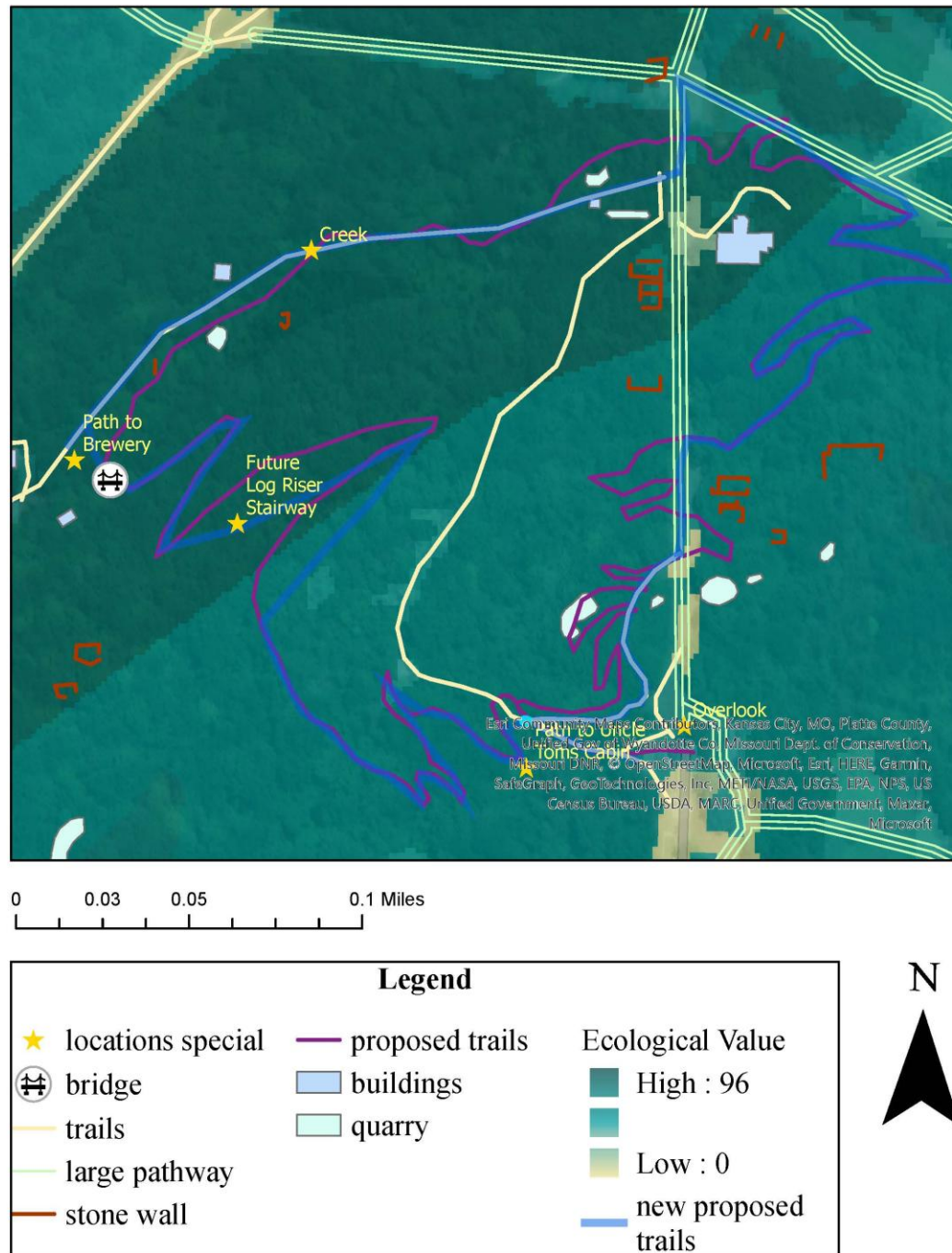


Figure 9. Proposed Bike Trails at Quindaro Ruins (created by Kaitlyn Rawlings)

8. Next Steps

The following actions are recommended to advance the management goals outlined in this plan:

- **Vegetation Survey:** Conduct a comprehensive vegetation survey of the Quindaro Ruins grounds. This will document current species composition, confirm the extent and distribution of invasive honeysuckle, and provide data to refine this plan.
- **Revegetation Plan:** Develop a detailed revegetation plan specifying trees, understory plants, and shrubs appropriate for post-removal planting. Prioritize native Kansas species; consider aesthetically impactful flowering natives near park trails for interpretive signage.
- **Structural Stabilization Plan:** Develop a detailed stabilization plan for archaeological features rated Poor in either Depositional Integrity or Conditions, in consultation with the National Park Service.
- **Stakeholder Coordination:** Establish a working group including the AME Church (landowner), Quindaro Ruins Project Foundation, NPS, KFS, Deep Roots KC, and Friends of Kaw Point Park to coordinate implementation.
- **Volunteer Program:** Build on existing volunteer efforts (e.g., JCCC-organized honeysuckle removal) by developing a structured, recurring volunteer program.

Appendix A: Resources

Below are some resources and links for more information on certain aspects of this land management document.

Site Features

NPS Assessment of Site Features:

<https://documentcloud.adobe.com/spodintegration/index.html?locale=en-us>

Floodplains

<https://www.americanrivers.org/threats-solutions/restoring-damaged-rivers/benefits-of-restoring-floodplains/>

<https://www.heartlandconservationalliance.org/blueriverreportcard>

Honeysuckle Removal

Economical Control: <https://bookstore.ksre.ksu.edu/pubs/MF3222.pdf>

Asian Bush Honeysuckle Removal in Kansas City:

https://www.kansasforests.org/forest_health/health_docs/Bush%20Honeysuckle%20One%20Pager.pdf

Guidance for Herbicide Use in Mistblowers:

https://www.kansasforests.org/forest_health/health_docs/Guidance%20for%20Herbicide%20Use%20for%20Mistblowers.pdf

Goats: <https://lawrenceks.org/2021/07/22/goats-to-become-the-g-o-a-t-of-eliminating-honeysuckle/>

Vegetation Removal

Managing archaeological evidence in woodland following tree removal:

<https://www.forestresearch.gov.uk/tools-and-resources/fthr/historic-environment-resources/managing-archaeological-evidence-in-wooded-environments/managing-archaeological-evidence-in-woodland-following-tree-removal/>

Revegetation For Archeological Site Stabilization:

https://www.nps.gov/subjects/archeology/upload/TchBrf8_508.pdf

Trails

Case Studies of Edge Effects:

- <https://scholarworks.montana.edu/xmlui/bitstream/handle/1/1236/EsbyE0811.pdf?sequence=1>
- https://wp.stolaf.edu/naturallands/files/2015/11/Butz_1999.pdf

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